ANUJ GUPTA

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1 Education

Indian Institute of Information Technology Senapati — Aug 2023 – Aug 2027
B.Tech in Electronics and Communication Engineering — *CGPA: 8.29 / 10*

- **Relevant Coursework:** Signals and Systems, Digital Signal Processing, Embedded Systems, Verilog HDL, Digital Design, Machine Learning Fundamentals, Control System, Data Science, Linear Algebra, Probability and Statistics

2 Experience

CodexIntern — Aug 2025 – Oct 2025
AI/ML Virtual Internship — *Remote*

- Worked on machine learning projects involving classification, regression, and NLP tasks.
- Performed data preprocessing, feature engineering, and model evaluation.
- Implemented supervised learning algorithms using Python and Scikit-learn.

MyCaptain — Jan 2024 – Mar 2024
Campus Ambassador — *Remote*

- Promoted technology bootcamps and learning programs among students.
- Guided and onboarded 30+ students to participate in skill-development programs.

3 Projects

AI-powered GATE Assistant with RAG | Streamlit, Gemini API, Python | Weblink | GitHub — 2025

- Built an AI-powered assistant for GATE aspirants supporting interactive Q&A and concept explanations.
- Implemented file upload for PDF/TXT/DOCX to extract and answer academic queries.
- Designed a Streamlit interface with downloadable PDF answers.
- Currently integrating Retrieval-Augmented Generation (RAG) for contextual and accurate responses.

Predicting Life Expectancy using Machine Learning | Python, Scikit-learn, Pandas, NumPy | GitHub — 2025

- Developed a machine learning model to predict life expectancy using global health and socio-economic indicators.
- Performed data preprocessing, feature selection, and exploratory data analysis on healthcare datasets.
- Implemented regression models to analyze relationships between factors such as healthcare expenditure, mortality rate, and life expectancy.
- Visualized insights using Matplotlib to interpret trends and model predictions.

High Performance Carry Save Adder (CSA) Design | Verilog, VLSI Design | GitHub — 2026 – Present

- Designing a 16-bit Carry Save Adder using parallel addition techniques to enable high-speed multi-operand arithmetic operations.
- Implemented RTL design in Verilog and performed simulation to verify functional correctness.
- Analyzing performance parameters such as latency, propagation delay, throughput, and power consumption.
- Planned synthesis and physical design flow using Cadence tools including Genus and Innovus for VLSI implementation.

SMS Spam Detection using NLP and Naïve Bayes | Python, Scikit-learn, NLP | Live | GitHub — 2025

- Developed a spam detection system using Naïve Bayes classification.
- Applied NLP preprocessing techniques and TF-IDF vectorization.

4 Ongoing Projects

MISTICS – Electrostatic Fog Collection System | *Electrostatics, Circuit Design, Sustainable Engineering* 2026

- Designing an electrostatic fog collection system to harvest water from atmospheric fog while reducing smog pollution.
- Applying electrostatic principles to enhance particle capture efficiency and improve environmental sustainability.
- Developing energy-efficient electronic circuits for optimized power consumption and continuous operation.

MEERA – Medical and Elderly Empowerment Residential Assistant | *IoT, Embedded Systems, Mesh Networking* 2026

- Developing a mesh-network-based home automation system to support healthcare monitoring and elderly assistance.
- Integrating hardware, firmware, and wireless communication for reliable device interoperability.
- Designing a decentralized architecture enabling energy-efficient operation without continuous internet dependency.

Network Intrusion Detection System using Machine Learning | *Python, Scikit-learn, Streamlit* 2026 – Present

- Developing a machine learning model to detect malicious network traffic using the NSL-KDD dataset.
- Implementing a Random Forest classifier to classify network activity and identify potential attacks.
- Performing data preprocessing, feature engineering, and model evaluation to improve detection accuracy.
- Building a Streamlit web interface for real-time intrusion detection and prediction.

5 Research Interests

Artificial Intelligence, Machine Learning, Internet of Things (IoT), Embedded Systems, Signal Processing, Sustainable Environmental Technologies

6 Technical Skills

Languages: Python, C, C++, Verilog

Machine Learning: Scikit-learn, Pandas, NumPy, Matplotlib

VLSI & Digital Design: Verilog HDL, Carry Save Adder Design, RTL Design

AI Technologies: Generative AI, Retrieval-Augmented Generation (RAG)

IoT & Embedded: Sensors, Microcontrollers, IoT Fundamentals

Web Development: HTML, CSS, Streamlit, Render

Tools: Git, VS Code, GNU Octave

7 Certifications

- Drone Bootcamp — Basics of drone technology and control
- Scilab Workshop — System simulation using Scilab
- IoT and Autonomous Systems Workshop — Covered IoT architecture, sensor integration, and autonomous system fundamentals